

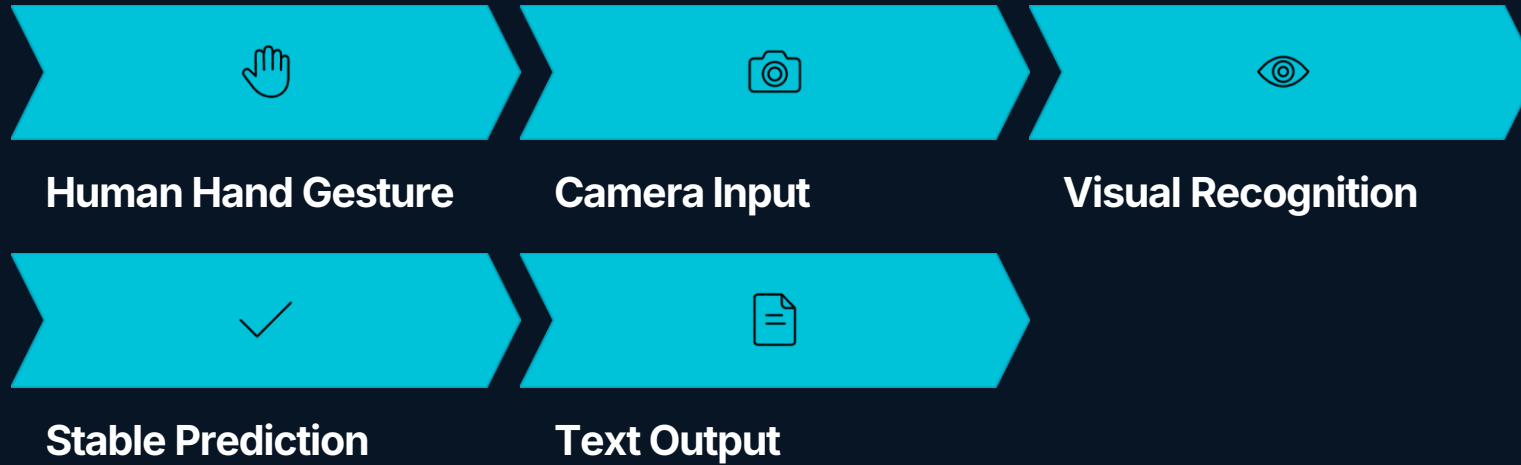
SignBridge

Real-Time American Sign Language Alphabet
Recognition

Computer Vision • Deep Learning • Hand Landmark Analysis • Real-Time Interaction

From gesture → recognition → text

The Problem: From Hand Gesture to Meaningful Text



Key challenges:

- Hand localization
- Similar-looking gestures
- Frame-to-frame flicker
- Dynamic signs
- Real-time interaction

📋 Build a real-time webcam system that recognizes static ASL alphabet gestures and turns stable predictions into text.

Detect Hand

Classify Gesture

Stabilize Prediction

Construct Text

Data & Deep Learning Pipeline

87,000 images

29 classes

A–Z, Space, Delete, Nothing

224 × 224 input

Model resolution

70 / 15 / 15 split

Train / validation / test

1

Dataset

2

Preprocessing & Augmentation

3

MobileNetV2
(ImageNet-pretrained)

4

Global Average Pooling

5

Dropout

6

29-Class Softmax

Transfer learning with ImageNet-pretrained MobileNetV2

- Augmentation: rotation, zoom, translation
- No horizontal flip — hand orientation matters

How SignBridge Works

PERCEPTION

Webcam • OpenCV • MediaPipe
Hand landmarks + preprocessing

INTELLIGENCE

MobileNetV2
Confidence filtering • landmark rules
Temporal smoothing • cooldown

INTERACTION

Word Builder
Streamlit output

1

Capture

2

Classify

3

Refine

4

Display

Making Recognition Stable in Real Time

Raw Frames

Temporal Vote



CONFIDENCE FILTERING

Reject low-confidence predictions



LANDMARK RULE ENGINE

Resolves confusing gestures: A/Y, D/G, F/W, H/B, I/Y, M/E, T/E, U/B, V/K

Stable
Prediction

Word Builder



TEMPORAL SMOOTHING

Majority vote over 6 predictions reduces flicker



COOLDOWN LOGIC

Prevents repeated insertion while a gesture is held

Output: A



Recognition combines classification, temporal smoothing, and interaction state.

Results, Applications & Current Boundaries

>90%

Validation accuracy documented in the project workflow

Documented validation result; final test-set metrics require reproduction of the selected checkpoint.

- Real-time webcam recognition
- 29-class prediction
- Confidence display
- Word construction

Applications



ASL Learning & Education



Accessibility Demonstrations



HCI Research

Limitations

- Static alphabet only
- J and Z need motion
- Single-hand runtime
- No continuous translation
- Sensitive to lighting and signer variation

From Static Gestures to Continuous Sign Understanding

SignBridge bridges webcam-based gesture recognition toward real-time sign-language interaction.

A modular foundation for accessible, real-time sign-language interaction.

CURRENT	NEXT PHASE	ADVANCED
Static ASL Alphabet	Dynamic Sign Recognition Two-Hand Recognition	Temporal Transformer / LSTM Language-Aware Decoding
DEPLOYMENT		
Speech Output Edge / Mobile Deployment		

Thank you — questions?